Experiment 1 – Optical free-beam transmission system

Preparation: comprehension questions, tasks and measurement plan with worked answers

Name, Group: Beyza Karatı, Saifeldin Ayman Elmoselhy Elnkep – Group 9 Date: 20.01.2027
(Antestat 14:00, building LT, room LT 227)

Quoted from the lab manual *Optoelectronics Lab: Optical free-beam transmission system* (e_poe_exp1.pdf): nine comprehension questions (§2–§4) and the practical tasks a)–g) (§6). Quotes are in shaded boxes with a teal bar; answers follow. Every answer ends with a short Turkish summary (**TR**). The manual asks that the questions be discussed as part of the preparation.

0. Background in one page

- **Optical transmission system:** electro-optical converter (LED / laser diode) → transmission path (free space or glass fibre, < 0.2 dB/km at 1.3 µm) → opto-electrical converter (photodiode). Here: LED – 1…80 cm of air – photodiode.
- **pn junction:** $I = I_S(e^{U/U_T} - 1)$, $U_T = kT/e \approx 26$ mV; space-charge zone, diffusion voltage $U_D \approx W_G/e$; forward: injection, reverse: tiny saturation current.
- **Photon energy / wavelength:** $E = h\nu$; $\lambda[\mu m] = 1.24 / E[eV]$; absorption for $E \geq W_G$ (generation of e-h pairs), radiative recombination for emission (spontaneous in the LED, stimulated in the laser).
- **Photodiode:** the field of the space-charge zone separates optically generated pairs → photocurrent $I_{ph} = S \cdot P$ (S = responsivity in A/W) even at zero voltage; in reverse bias linear and fast.
- **System equation:** $I_A = \eta_E \cdot \kappa \cdot \eta_A \cdot I_E$ (LED conversion × path transmission × PD conversion).
- **Radiation pattern:** LED with diffuse coloured housing ≈ Lambert $I = I_0 \cos\theta$ (beam angle 60°); transparent housing with lens dome ≈ 12°. Beam angle = angle where the intensity falls to ½.
- **Dynamic behaviour:** the link behaves like an RC low-pass: t_r (10–90 %), $f_{gr} = 0.35/t_r$, f_{3dB}, phase shift / signal delay t_0.

Questions §2 – the pn transition

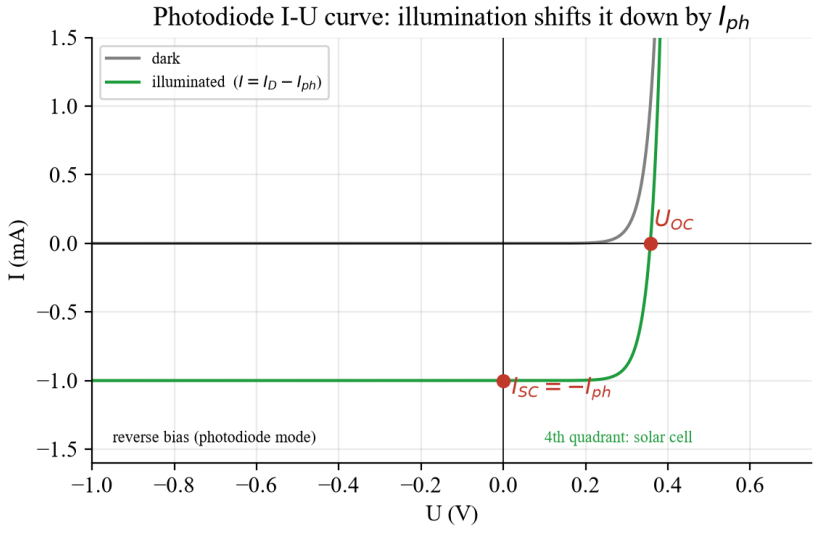


Fig. A – Photodiode I-U curve (model): dark curve and illuminated curve shifted down by I_{ph}.

Question 1 (§2): Which properties of the pn junction result from the band diagram and the characteristic curves?

- **Band diagram:** equalised Fermi levels → space-charge zone depleted of free carriers with a built-in field and a **diffusion voltage U_D** ($\approx W_G/e$); p-region holes, n-region electrons.

- **Characteristic curve:** a **rectifier**: forward current rises exponentially above ≈ U_D (injection of minority carriers, recombination → light in the LED), reverse current only the small saturation current I_S (generation in the zone; photocurrent adds here).
- **Further:** temperature dependence ($I_S \propto e^{-W_G/kT}$, U falls ≈ 2 mV/K), junction capacitance (limits speed), breakdown in reverse direction, ideality factor n = 1…2.

TR Bant diyagramı: uzay yükü bölgesi, difüzyon gerilimi U_D ≈ W_G/e. Eğri: doğrultucu; ileri yönde üstel, ters yönde I_S; sıcaklık, kapasite, çığ kırılması.

Question 2 (§2): How does the characteristic curve shown in Fig. 3 change when the photodiode is illuminated?

It is **shifted down along the current axis by the photocurrent**: $I = I_S(e^{U/U_T} - 1) - I_{ph}$ (superposition, Fig. A). Consequences: at U = 0 the short-circuit current is $I_{SC} = -I_{ph}$; the curve crosses the voltage axis at the open-circuit voltage $U_{OC} = U_T \ln(I_{ph}/I_S + 1)$ (4th quadrant = solar-cell operation); in reverse bias the current is simply $-I_{ph}$, proportional to the optical power.

TR Aydınlatınca eğri I_ph kadar aşağı kayar: I = I_S(e^(U/U_T)−1) − I_ph; U=0'da I_SC = −I_ph, açık devrede U_OC = U_T ln(I_ph/I_S+1).

Question 3 (§2): When is there a linear relationship between the optical power P and the electrical measured variable I?

- **Photodiode:** $I_{ph} = (\eta\, e/h\nu)\cdot P$ is linear when the diode is in **reverse bias or short circuit** (load resistance so small that the voltage across the diode stays ≈ 0 or reverse) and the optical power is not so large that the carriers screen the field (saturation).
- **With a load resistor R** (20 kΩ in this experiment): $U = R\cdot I_{PD}$ grows with P; at large P the PD is pushed towards forward bias (U_{OC} log-limit) and the relation becomes sub-linear – use small R or low power.
- **LED:** $P \propto I$ (η_{ext} constant) at moderate currents; at high currents heating/droop makes it sub-linear.

TR Fotodiyot: ters gerilim/kısa devre ve doymayan güçlerde I_ph = S·P doğrusal. Yük direnci büyükse veya güç çok yüksekse doyma. LED: orta akımlarda P ∝ I.

Question 4 (§2): How can the transmission properties of the components (LED, photodiode) be determined?

- **LED:** conversion efficiency $\eta_E = \Phi_E/I_E$ – measure the emitted power (calibrated power meter / reference PD) versus current; or electrical/optical: U-I curve for U_D, P(I).
- **Photodiode:** responsivity $S = I_{ph}/P_{opt}$ in A/W with a known optical power (reference detector); linearity from I_{ph} versus P (here: distance r gives $P \propto 1/r^2$).
- **Dynamic:** rise and fall time with a square wave, frequency response (amplitude, phase/delay) with a sinusoidal signal, f_{3dB} – tasks d)–f).

TR LED: P–I (η_E); PD: S = I_ph/P (A/W); dinamik: yükselme/düşme süresi, genlik–faz–frekans eğrisi, f_3dB.

Questions §3 – the optical transmission system

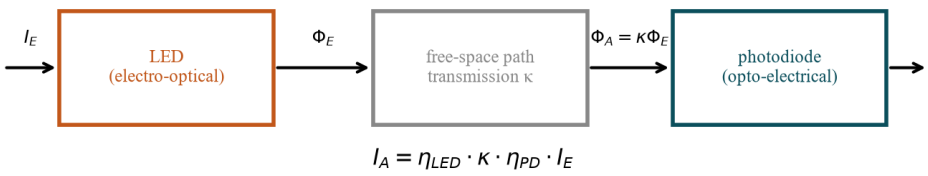


Fig. B – Block diagram of the transmission system.

Question 1 (§3): How is the transmission characteristic of the overall system determined from the individual components?

From the chain: LED (η_E: $\Phi_E = \eta_E\, I_E$) → path ($\Phi_A = \kappa\, \Phi_E$) → PD ($I_A = \eta_A\, \Phi_A$):

$$I_A = \eta_{LED} \cdot \kappa \cdot \eta_{PD} \cdot I_E \qquad \text{(in dB: the individual gains/losses add)}$$

TR Zincir: I_A = η_LED·κ·η_PD·I_E; dB cinsinden halkalar toplanır.

Question 2 (§3): When is the system a linear, time-invariant system and how can this be verified experimentally?

- **Linear:** superposition – response to $a \cdot x_1 + b \cdot x_2$ is $a \cdot y_1 + b \cdot y_2$: LED in its linear P-I region (bias keeps it on), PD in reverse/short circuit, no saturation, no clipping.
- **Time-invariant:** the parameters (η, κ, temperature, path) do not change in time: a time-shifted input gives the same, equally shifted output.
- **Verify:** a sine input gives a sine of the same frequency (no harmonics); double the amplitude → double output; sum of two frequencies → sum of the responses; repeat the measurement after a time delay → identical curve.

TR Doğrusal: süperpozisyon (genlik ×2 → çıkış ×2, sinüs → aynı frekanslı sinüs); zamanla değişmeyen: parametreler sabit (zaman kaydır → aynı kayma).

Question 3 (§3): What changes if the length of the transmission path is changed?

- **κ changes:** the radiation spreads over a larger area: far from the LED (r much larger than the component size) the irradiance ∝ 1/r², so $I_{PD} \propto 1/r^2$ (−20 dB per decade in I, −40 dB in electrical power I²R). In this experiment r is only 0 … 2.5 cm (comparable to the 5 mm components), so the near-field law $I_{PD} = I_0/(1 + r/r_0)^2$ with an effective source distance r_0 describes it better (see task b).
- **Delay:** t = r/c ≈ 3.3 ns per metre – negligible against the µs time constants here.
- **Bandwidth and linearity of the components do not change** – only the overall gain I_A/I_E (the system stays linear).

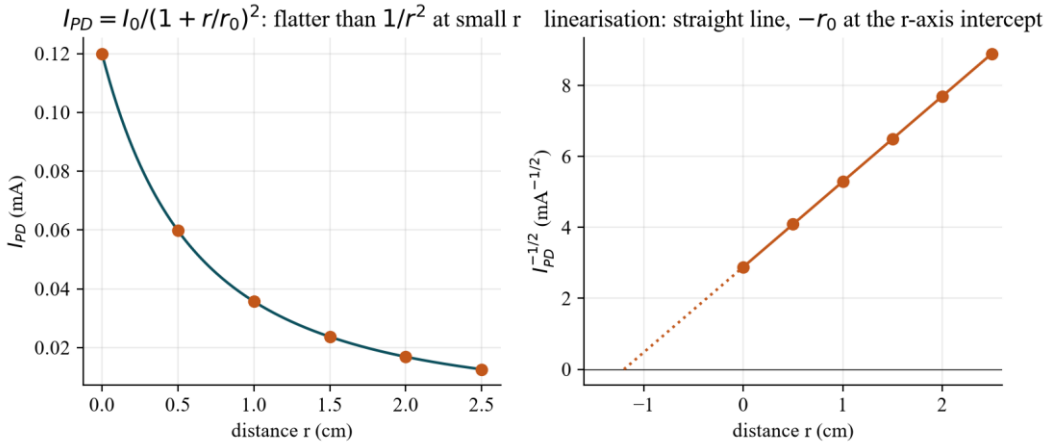


Fig. C – Model for the template distances r = 0 … 2.5 cm: $I_{PD} = I_0/(1 + r/r_0)^2$; plotting $I_{PD}^{-1/2}$ versus r gives a straight line whose r-axis intercept is $-r_0$.

TR Mesafe değişince yalnızca κ değişir: uzak alanda I_PD ∝ 1/r²; bu deneyde r = 0…2,5 cm (bileşen boyutu mertebesi) → I = I_0/(1+r/r_0)²; I^(−1/2)–r doğrusu. Gecikme r/c ihmal edilebilir; sistem doğrusal kalır.

Questions §4 – radiation properties

Radiation of two LEDs (Fig. 7 of the manual)

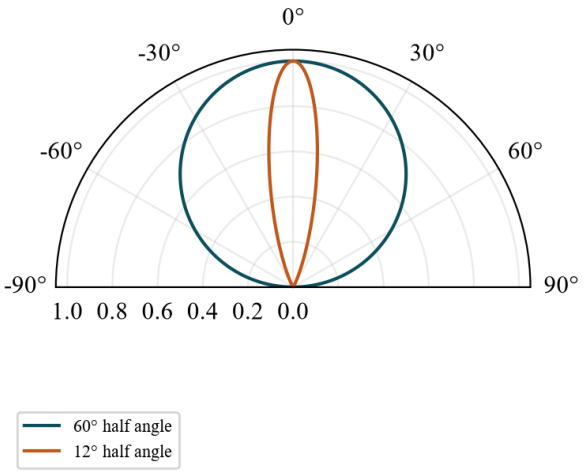


Fig. D – The two LED patterns of the manual: Lambert (beam angle 60°) and a lens LED (12°).

Question 1 (§4): Where can both LEDs be used?

- **60° (Lambert, diffuse coloured housing):** indicator and signal lamps, displays, backlighting, short-distance links where the receiver position is not exactly known – wide coverage, tolerant to misalignment.
- **12° (clear lens housing):** free-space links over longer distance, light barriers, remote controls, sensors, illuminating a spot – much higher intensity on the axis for the same power, but alignment is critical.

Question 2 (§4): What designs do they have?

- **60°:** coloured, diffusely scattering plastic housing (scattering centres), roughly flat/low dome, no focusing.
- **12°:** highly transparent uncoloured housing with a lens-shaped dome (and often a reflector cup around the chip) that collimates the light; the chip sits near the focus.

Why the on-axis intensity differs: the same total power is concentrated into a much smaller solid angle ($\Omega \approx \pi\cdot(\sin(\varphi))^2$); 12° versus a hemisphere-like pattern gives roughly a factor 10–25 higher intensity on the axis, so the 12° LED delivers a much larger I_{PD} at the same distance r.

TR 60°: difüz renkli gövde, geniş alan (gösterge, kısa menzil, hizaya toleranslı); 12°: şeffaf mercekli kubbe, dar ışın (serbest uzay hattı, bariyer, kumanda), eksen yoğunluğu yaklaşık 10–25 kat yüksek.

Practical tasks §6 – what to prepare

Equipment: LED and photodiode on a rail (distance r adjustable), two multimeters (HP 3478A for current, Kontron for voltage), voltage source (U_{Netz} = 7 V for the LED), Keysight EDUX1022G oscilloscope with function generator. The protocol template prescribes the measuring points (quoted below).

a) LED I-U characteristic

Task a): The current-voltage characteristic of the LED must be determined. Use a digital multimeter (Hewlett Packard) to determine the current through the diode and a second digital multimeter (Kontron) to measure the voltage across the LED. Apply: I(U). Determine: U_D.

Template table: you set the source voltage $\mathbf{U_{Quelle}}$ **= −1.0, −0.5, 0, 0.5, 1.0, 1.2, 1.4, 1.6, 1.8, 2.0, 2.5, 3.0, 3.5, 4.0, 4.5, 5.0, 6.0, 7.0 V** (fine steps at the knee) and read $\mathbf{U_{LED}}$ (Kontron multimeter across the LED) and $\mathbf{I_{LED}}$ (HP multimeter in series); plot $I_{LED}(U_{LED})$. **Forward voltage U_F (= U_D):** the voltage at which the current becomes noticeable – state a clear definition, e.g. U_{LED} at 1 mA, or the intercept of the linear

extrapolation of the steep part with the voltage axis. Expectation $U_F \approx W_G/e$: IR ≈ 1.2 V, red ≈ 1.8–2.0 V, green ≈ 2.1 V, blue ≈ 3 V. In reverse direction ($U_{Quelle} < 0$) the current is ≈ 0 – the LED is a diode. ($U_{Quelle} - U_{LED}$)/I_{LED} gives the series resistance of the test circuit – a useful cross-check; it also limits the current at 7 V.

b) PD – the I(P_PD) characteristic

Task b): The I(P_{PD}) characteristic curve of the PD is to be recorded via the proportionality of the distance r between the LED and PD and the power P_{PD} generated at the photodiode. The load resistance at the PD is R = 20 kΩ. To do this, the voltage across the load resistor R: $I_{PD} = U/R$ must be recorded with the HP digital multimeter and the distance r must be changed appropriately. The voltage at the power supply unit for the LED should be set to $U_{Netz} = 7$ V. Apply: I(r), $I_{PD}(P_{PD})$ (P_{PD} in W and dBm).

Template tables: distance **r = 0, 0.5, 1.0, 1.5, 2.0, 2.5 cm** (0 = smallest adjustable distance); measure $\mathbf{U_{PD}}$ across the 20 kΩ resistor (HP multimeter) and compute $\mathbf{I_{PD} = U_{PD}/R}$ (mA), $\mathbf{P_{PD} = U_{PD}\cdot I_{PD} = U_{PD}^2/R}$ (W) and $\mathbf{P_{PD}}$ **in dBm =** $\mathbf{10\ log_{10}(P/1\ mW)}$; diagrams: $I_{PD}(r)$, $P_{PD}(I_{PD})$ in W and in dBm. Example: U_{PD} = 2.0 V → I = 0.10 mA, P = 0.20 mW = **−7 dBm**; U = 0.4 V → I = 20 µA, P = 8 µW = −21 dBm.

What to expect: the optical power at the PD falls with r (far-field law $\propto 1/r^2$). With r ≤ 2.5 cm – comparable with the 5 mm size of LED and PD – the point-source law does not hold from r = 0; use the effective-distance form $\mathbf{I_{PD} = I_0/(1 + r/r_0)^2}$ (Fig. C): the plot of $I_{PD}^{-1/2}$ against r is a straight line with the r-axis intercept at $-r_0$ (distance from the emitting chip/lens to the effective source plane). **Linearity of the PD:** $I_{PD} \propto$ optical power as long as the PD is not saturated; with R = 20 kΩ the voltage U_{PD} may reach several volts at r = 0 and the diode leaves the reverse/short-circuit regime → sub-linear at the smallest r.

TR Şablon: r = 0, 0,5 … 2,5 cm; U_PD ölç → I_PD = U/R (R = 20 kΩ), P = U²/R, dBm = 10 log(P/1 mW). I^(−1/2)–r doğrusu r_0'ı verir; r = 0'da PD doyabilir.

c) Overall efficiency

Task c): For the smallest adjustable distance r_{min}, the electrical power consumed by the LED P_{LED} (U_{Netz} = 7 V, I_{LED} = ?) and the electrical power emitted by the photodiode P_{PD} (R = 20 kΩ, I_{PD} = ?) should be determined. Determine: P_{LED}, $P_{PD}(r_{min})$, $\eta(r_{min})$.

$$P_{LED} = U_{Netz} \cdot I_{LED} \qquad P_{PD} = I_{PD}^2 \cdot R \qquad \eta(r_{min}) = P_{PD} / P_{LED}$$

Example (illustrative numbers): I_{LED} = 15 mA → P_{LED} = 7 V · 15 mA = 105 mW; I_{PD} = 20 µA → P_{PD} = (20 µA)²·20 kΩ = 8 µW; $\eta = 8\cdot10^{-6} / 0.105 \approx \mathbf{8\cdot10^{-5} \approx -41\ dB}$. Explain the tiny value: LED conversion (a fraction of the electrical power becomes light), geometric coupling κ (solid angle of the PD), PD conversion, and the electrical power in a high-ohmic load of 20 kΩ is far below the PD's possible output. $\mathbf{P_{LED} = U_{Netz}\cdot I_{LED}}$ **uses the supply voltage (7 V), i.e. it includes the loss in the series resistor.**

d) Rise and fall time, cut-off frequency

Task d): For these AC measurements, the signal generator (offset 5 V, amplitude 5 V, square-wave voltage) must be connected to the LED via the series resistor. The dual-channel oscilloscope should tap the voltage across the LED and series resistor on channel 1 and the voltage across the resistor on the PD on channel 2. Triggering takes place on channel 1. At a frequency of $\nu = 20$ kHz, the rise time t_{rise} and the fall time t_{fall} are to be determined. Provided that the system can be described by an RC low-pass filter, it is possible to determine the cut-off frequency f_{gr} using $f_{gr} = (\ln 0.9 - \ln 0.1)/(2\pi\cdot(t_{rise}+t_{fall})/2) \cong 0.35 / ((t_{rise}+t_{fall})/2)$.

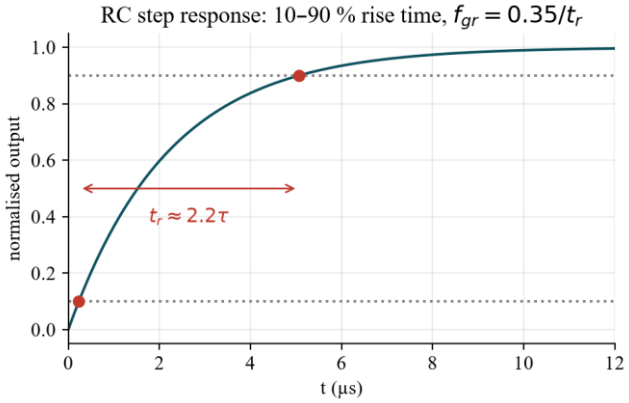


Fig. E – RC step response: the 10–90 % rise time is 2.2 τ.

RC low-pass: step response $y = 1 - e^{-t/\tau}$; the 10 % and 90 % points lie at $t = 0.105\tau$ and $2.30\tau \rightarrow \mathbf{t_r = 2.2\ \tau}$; $f_{gr} = 1/(2\pi\tau) = 2.2/(2\pi\ t_r) = \mathbf{0.35/t_r}$ (the manual's ln-expression is the same). Measure t_{rise} (10 % → 90 % of the final value) on the rising edge and t_{fall} (90 % → 10 %) on the falling edge of channel 2; use a fast time base so that the edge fills the screen, and check that the signal reaches its final value within the half period (20 kHz → 25 µs). *Example:* $t_r = 4.0$ µs, $t_f = 4.4$ µs → $f_{gr} = 0.35/4.2$ µs ≈ **83 kHz**.

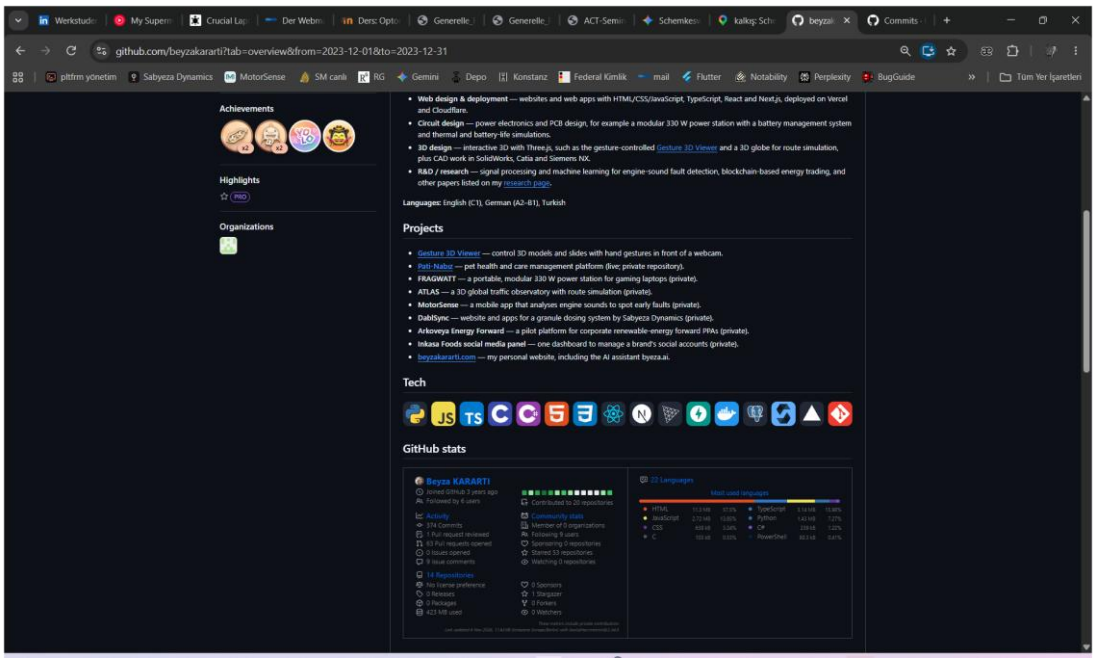


Fig. F – LTspice demo (optional): RC low-pass with R_s = 1k, 2.2k, 4.7k and C = 1 nF; t_{rise} = 2.2 / 4.8 / 10.3 µs, f_{gr} = 159 / 72 / 34 kHz.

TR RC: t_r(10–90 %) = 2,2 τ, f_gr = 1/(2πτ) = 0,35/t_r; örnek t_r = 4 µs, t_f = 4,4 µs → f_gr ≈ 83 kHz. Seri direnç büyürse τ=RC büyür, f_gr düşer.

e) Signal delay and phase shift

> **Task e):** The frequency range ν = 1 kHz … 80 kHz should be scanned with the signal generator (offset 5 V, amplitude 5 V, sinusoidal voltage) and the signal delay t_0 between channel 1 and channel 2 should be measured. Apply: t_0, φ(ν).

Template table (f in kHz): 1, 2, 5, 10, 20, 30, 40, 50, 60, 70, 80 → columns **t_0 (µs), φ (rad), φ (°)**. Measure the time shift t_0 between the same phase points (zero crossings, same slope) of channel 1 and 2; **φ[°] = 360° · f · t_0**, φ[rad] = 2π f t_0. For an RC low-pass φ = arctan(f/f_{gr}) (≈ 45° at f_{gr}, → 90° for f ≫ f_{gr}), hence t_0 = φ/(360° f): constant ≈ RC = 1/(2πf_{gr}) at low frequency, **falling ∝ 1/f** at high frequency (it is a low-pass, not a pure delay line). Plot t_0(f) and φ(f) (Fig. 5 of the template).

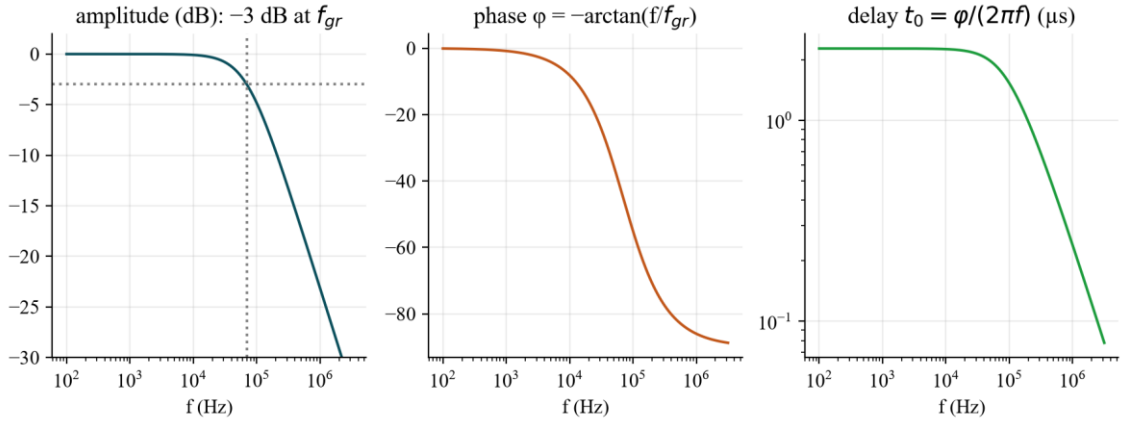


Fig. G – Model low-pass (f_{gr} = 70 kHz): amplitude, phase and delay versus frequency.

f) Cut-off frequency (−3 dB)

> **Task f):** The frequency must be increased at the signal generator (offset 5 V, amplitude 5 V, sinusoidal voltage) from ν = 250 Hz until the amplitude of the signal at the PD has fallen by 3 dB. Determine: cut-off frequency f_{3dB}.

Template table: f = 250, 500, 1000 Hz … (continue upward as needed) with columns **U_{PD} (V), P_{PD} (dBm), level (dB)**; the level is 0 dB at 250 Hz: **level = 20 log(U_{PD}(f)/U_{PD}(250 Hz))** (= P_{PD}[dBm](f) − P_{PD}[dBm](250 Hz)). Raise f until the level is **−3 dB** (amplitude 0.708 ×) → f_{3dB}; plot the **Bode diagram** (level in dB over f on a log axis, Fig. 6). Compare f_{3dB} with f_{gr} from the rise time (task d): for a first-order low-pass they agree; differences indicate non-ideal behaviour (LED/PD are not a single RC element, nonlinearities, oscilloscope bandwidth, probe capacitance).

g) Optical audio transmission

> **Task g):** Construct an optical free-beam transmission system with an LED as the signal source and a photodiode as the signal receiver and then transmit audio signals. The length of the free-beam path should be at least 1 m. (Protocol: describe the differences between analog and digital signal transmission.)

- **Set-up:** audio source → amplifier/offset (bias so that the LED current stays positive and in its linear region) → LED; 1 m free path, **aligned** (use the narrow 12° LED for more signal); photodiode (reverse biased) → amplifier/load → loudspeaker/headphones.
- **Analog:** the audio waveform modulates the LED intensity continuously; needs linearity (bias + small modulation index), noise and distortion add directly to the signal, quality degrades gradually with distance; simple and cheap.
- **Digital:** the signal is sampled and coded as bits (LED on/off); a threshold decision regenerates the signal, so noise below the threshold does no harm and errors can be corrected, but more bandwidth (rise time, f_{gr} ≫ sample rate) and converters (ADC/DAC) are needed; quality is constant until the error rate suddenly rises.
- **Practice:** ambient light (steady offset, 100 Hz hum from lamps) disturbs the PD – block it with AC coupling, shade the receiver, increase R_{load} for more signal (but lower bandwidth).

TR Analog: ses LED şiddetini sürekli modüle eder (doğrusallık, bias, gürültü doğrudan karışır); dijital: bit/eşik kararı, gürültüye dayanıklı ama daha geniş bant ve ADC/DAC gerekir. Ortam ışığı (100 Hz) AC kuplajla engellenir.

Antestat – likely questions with short answers

Question	Short answer
What are the three elements of an optical transmission system?	Electro-optical converter (LED/laser), transmission path (air/fibre), opto-electrical converter (photodiode).
What is the transmission equation of this system?	$I_A = \eta_{LED} \cdot \kappa \cdot \eta_{PD} \cdot I_E$.
How does the I-U curve of a photodiode change with light?	Shifted down by I_{ph}; $I_{SC} = -I_{ph}$, $U_{OC} = U_T \ln(I_{ph}/I_S + 1)$.
When is I proportional to P?	Photodiode in reverse/short circuit with small load and no saturation; LED at moderate currents.
How does I_{PD} depend on the distance?	Far field: $\propto 1/r^2$ (irradiance of a point-like source); at r = 0…2.5 cm: $I_0/(1 + r/r_0)^2$; electrical power $\propto I^2$, i.e. −40 dB per decade in the far field.
What is the beam angle?	The angle at which the intensity falls to half of its maximum: 60° (Lambert) vs. 12° (lens LED).
Why is the efficiency so small ($\approx 10^{-4}$)?	Conversion in the LED, geometric coupling κ and conversion in the PD multiply; the load of 20 kΩ extracts little electrical power.
What is the relation between rise time and cut-off frequency?	RC low-pass: $t_r = 2.2\,\tau$, $f_{gr} = 1/(2\pi\tau) = 0.35/t_r$.
What does the phase shift of the link look like?	$\varphi = \arctan(f/f_{gr})$: 45° at f_{gr}, 90° for $f \gg f_{gr}$; delay $t_0 = \varphi/(360°\, f)$.
How do you measure the −3 dB frequency?	Increase the sine frequency from 250 Hz until the PD amplitude has dropped to 0.708 of the low-frequency value.
Why offset 5 V and amplitude 5 V?	The LED conducts only in forward direction: 0…10 V keeps the LED current positive; the offset is the bias point.
Linear, time-invariant – how to check?	Sine in → sine out, double the amplitude → double output, shift in time → same shift.
Analog vs. digital transmission?	Analog: continuous intensity modulation, noise adds; digital: bits with threshold decision, robust but needs bandwidth and converters.

Lab-day checklist

- Antestat **14:00**, building **LT, room LT 227**, group 9, **20 January 2027**.
- **Bring** (manual §7): calculator, pencil, **30-cm ruler, linearly divided graph paper**; this preparation; the protocol template (it prescribes the measuring points).
- The written elaboration (graphs, explanations) is done after the lab; the tasks a)–g) must be finished during the lab – prepare the tables with the template's points.
- a) $U_{Quelle} = -1$ … 7 V with fine steps at 1.2–1.8 V (read U_{LED}, I_{LED}); b) r = 0…2.5 cm in 0.5 cm steps with U_{Netz} = 7 V and R = 20 kΩ; c) $P_{LED} = 7\ V \cdot I_{LED}$ and $P_{PD} = I_{PD}^2 R$ at r_{min}; d) 20 kHz square (offset 5 V, amplitude 5 V), 10–90 %; e) f = 1, 2, 5, 10 … 80 kHz delay and phase; f) 250, 500, 1000 Hz … −3 dB; g) ≥ 1 m free path.
- Note the LED type (colour, beam angle), the series resistor value, and all cursor/time-base settings.